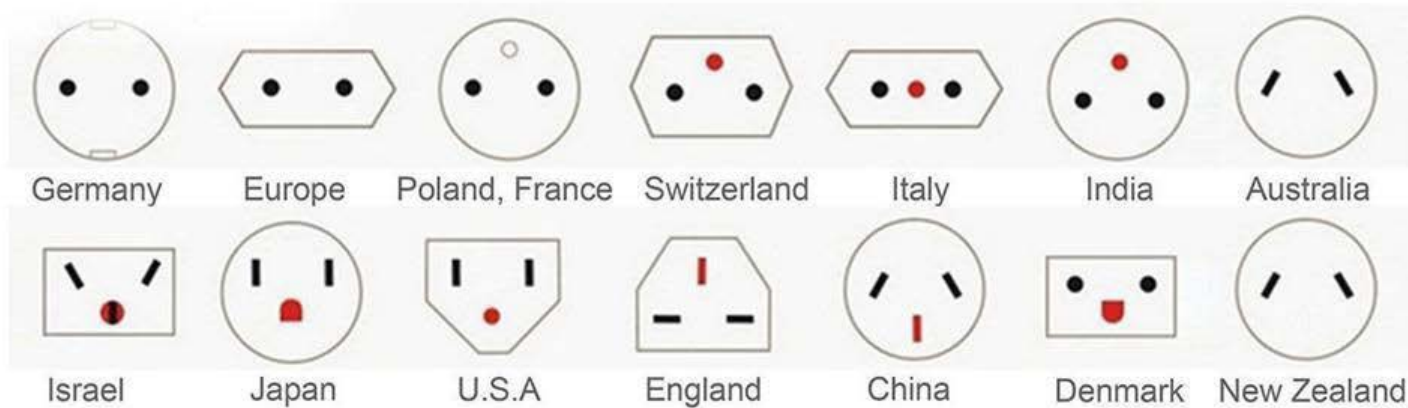


Prof. Dr.-Ing. Viktor Wesselak

Technical Standards

Lecture Scientific Practice

Why technical standards?

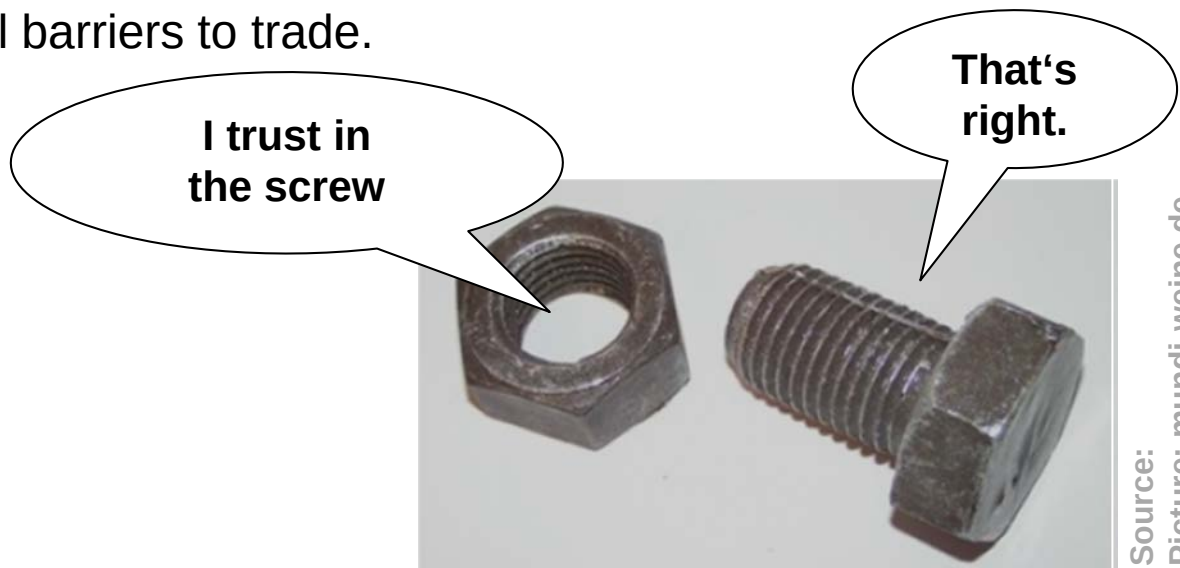


Definition

„Technical standards are one-off solutions to repetitive tasks published by a standards organisation“.

➔ Standards aim at standardisation by selecting a specific one from several possible technical solutions.

- Products of different companies fit together.
- No explicit coordination is necessary.
- Reduction of technical barriers to trade.



Definition

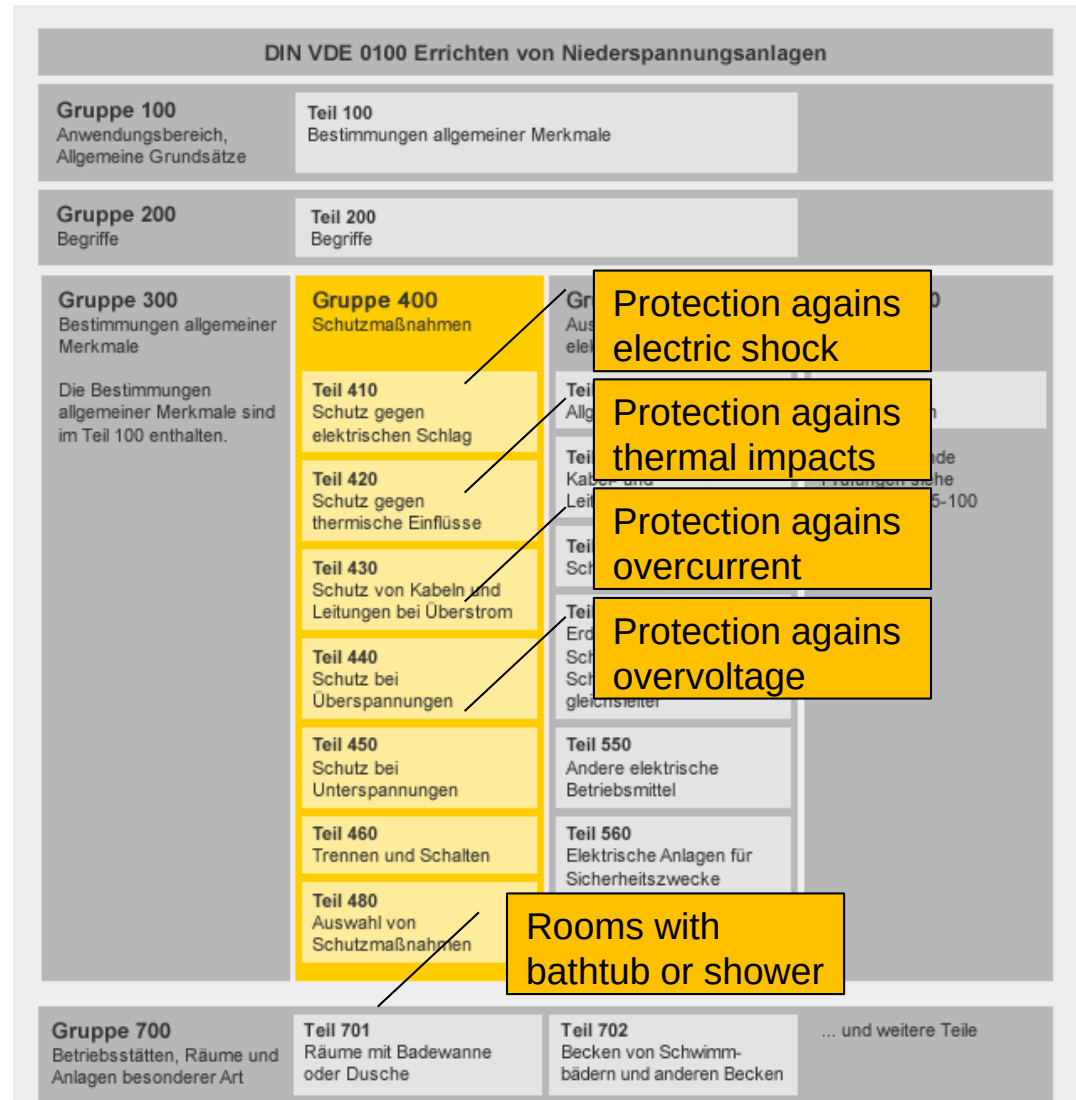
- ➔ Standards help a company, as it does not have to come up with new ideas for solved technical problems.
- Standards reflect the state of the art.
 - They define a certain level of requirements, e.g. regarding the safety of technical solutions. ("Compliance with the generally recognised codes of practice")

Example:: DIN VDE 0100
"Installation of low voltage systems"



Example

Strukture of the
DIN VDE 0100



Principles of standardisation (DIN 820)

"Standardisation is the planned unification of material and immaterial opposites for the benefit of the general public, carried out jointly by relevant stakeholders. It must not lead to a special economic advantage for private parties".

Voluntarity

Participation in standardisation and its use is basically optional.

- Standards are not laws or regulations; each company can decide for itself whether to manufacture its products according to the relevant standard or to go its own way.
- However, the compliance of technical standards is often economically unavoidable, otherwise there is a risk of exclusion from markets, e.g. quality management system according to ISO 9000

Principles of standardisation (DIN 820)

The Public

- All standardisation projects are publicly announced and submitted to the public for comments (www.din.de).

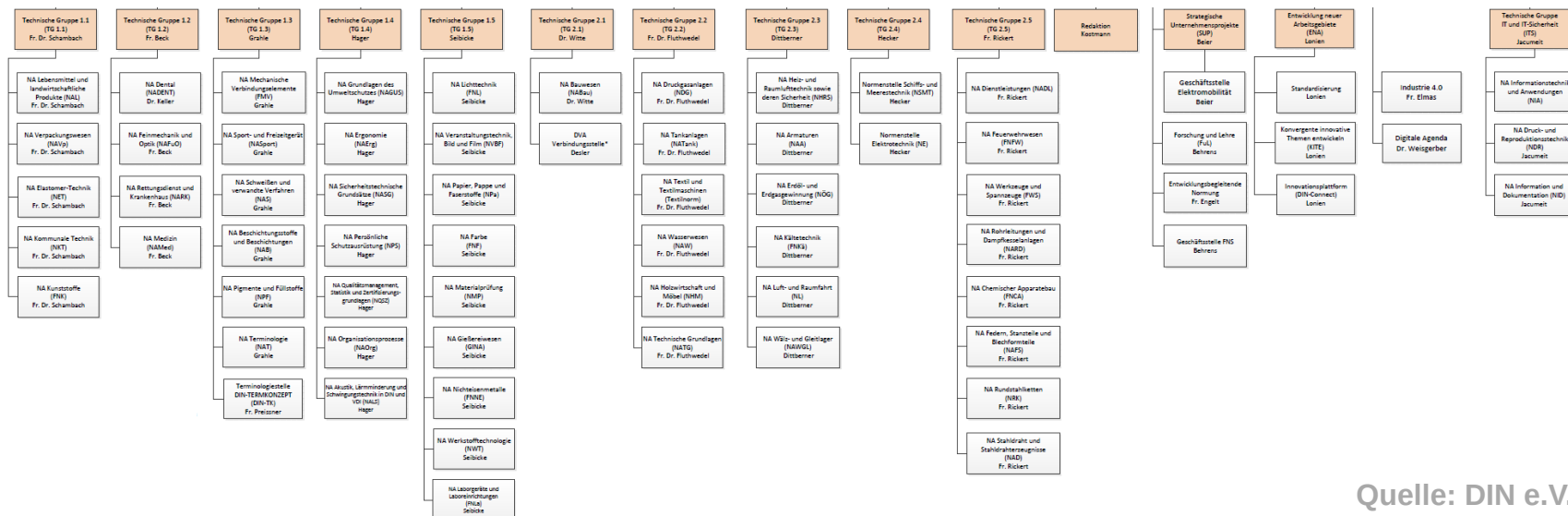
Participation

- All German standardization projects are discussed in a total of more than 70 standards committees by experts from the "relevant stakeholders".
- The "relevant stakeholders" include manufacturers, trade, consumers, science, test institutes, craftsmen.
- The work in the standards committees should lead to a consensus supported by all stakeholders.
- Standards are based on the state of the art and is therefore reviewed regularly (e.g. every 5 years).

Organisation of standardisation

Germany

- In Germany, the standardisation is carried out by DIN e.V. (Deutsches Institut für Normung) and the DKE (Deutsche Kommission Elektrotechnik Elektronik Informationstechnik)).
- DIN e.V. publishes the DIN standards through its publishing house: www.beuth.de
- The standards committees organised by DIN e.V:



Organisation of standardisation

Europe

- In Europe, standardisation work is organised by three institutions:
 - the European Institute for Standardisation (CEN)
 - European Committee for Electrotechnical Standardization (CENELEC)
 - European Telecommunications Standards Institute (ETSI)
- Germany is represented by the DIN e.V. in the European standards committees ("mirror committees").
- Decisions are taken according to the majority principle. In most cases, one member per country, but voting weight is based on economic strength.
- European standards must be transferred into national standards.

Organisation of standardisation

Global

- Standardisation work is organised internationally by three institutions:
 - International Standardisation Organisation (ISO)
 - International Electrotechnical Commission (IEC)
 - International Telecommunications Union (ITU)
- Germany is represented by the DIN e.V. in the international standardisation organisations.
- Decisions are made according to the majority principle. The following applies: one vote per country.
- International standards can be transferred to the national set of standards. The adoption is usually indicated by the nomenclature: DIN EN ISO xxx